

Angel Fernando Bórquez Guerrero

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Hermosillo, Sonora.

Education

Universidad de Sonora
Bachelor of Science in Computer Science

CGPA: 91.91/100
August 2023 – Present

Research Experience

University of Guadalajara

Undergraduate Research Intern (Advisor: Dr. Rafael Morales Gamboa)

Guadalajara, México
June 2025 – July 2025

- Validated and generalized an Entropic Associative Memory (EAM) model across high-dimensional benchmark datasets, culminating in a peer-reviewed publication at *IEEE ENC 2025*.
- Designed an experimental novelty detection framework to rigorously evaluate the EAM architecture's out-of-distribution (OOD) class rejection performance under varying operational bounds.
- Coupled deep perceptual networks (convolutional autoencoders and classifiers) to project raw image spaces into bounded latent vector representations compatible with associative memory topologies.
- Formulated quantitative control protocols using specialized “no-response” metrics to model partial memory saturation and characterize systematic false-positive rejection dynamics.
- Engineered an end-to-end reproducible simulation framework, modularizing class density distributions, balance ratios, and multi-class retrieval evaluation regimes.

Publications

- A. F. Bórquez Guerrero**, R. Morales Gamboa, et al. “*Large-Scale Validation and Analysis of the Rejection Capacity in Entropic Associative Memory*.” In **Proceedings of the 2025 Mexican International Conference on Computer Science (ENC) & IEEE ICEV**, Orizaba, Veracruz, México, 2025. IEEE.
Indexing: Online ISSN: 2332-5712 | ISBN: 979-8-3315-9026-0/25 | IEEE Xplore.

Professional Experience

TE Connectivity

AI Engineer Intern

Supervisor: M.Sc. Martín Vega
June 2026 – September 2026

- Designed and developed an AI agent for industrial alert analysis, implementing its orchestration, state management, tool integration, and execution workflows to support automated analysis and decision-making.
- Re-architected the agent's orchestration using LangGraph, replacing the previous workflow implementation with a graph-based architecture that improved state management, modularity, and execution efficiency.
- Optimized the agent's architecture and execution pipeline, reducing response times from up to three minutes to near-instant responses and substantially improving its practical usability.
- Designed and deployed a shared Linux development server for interns, configuring the system and user access to provide a centralized environment for developing and hosting research and engineering projects.

- Professional internship in the Supercomputing area at the University of Sonora.
- Part of a development team building a specialized forum for users of *Yuca*, the university HPC system.
- Implementing features such as user posts, direct messaging, and service request workflows.
- Supporting an active HPC user community by improving communication and technical support processes.

Competitions & Projects

TE AI CUP 2026 – International 1st Place

Project: Industrial Alert Agent
January 2026 - June 2026

Project Leader

- Led a 6-member interdisciplinary team in an international AI competition, guiding the technical development of an AI-driven system for industrial alert analysis that achieved 1st place.
- Developed an AI agent integrating Large Language Models (LLMs) for automated alert analysis and decision support.
- Developed an end-to-end data and machine learning pipeline involving data cleaning, text preprocessing, embedding generation, clustering, data validation, and predictive modeling.
- Developed a predictive modeling system to estimate the likelihood of future industrial alerts, providing a proactive component that complemented the analysis of historical patterns and enabled earlier identification of potential issues.
- Architected the project around clean, standardized, and scalable software structures, establishing conventions for modules, tools, configuration, state management, and component integration.

HSIL Hackathon 2026 – 3rd Place

Project: ShiftGuardian
April 2026

Application Designer & AI Agent Architect

- Designed the application interface and AI agent architecture for a clinical decision support system supporting nurse handoffs and patient prioritization.
- Integrated a RAG-based LLM workflow concept to summarize clinical reports, identify critical tasks, and reduce cognitive load during shift transitions.

MICAI 2025 MeDA Challenge – National 1st Place

October 2025

- Contributed to a 1st Place victory in a national academic competition focused on Medical Domain Adaptation using Self-Supervised Learning (SSL), leading to an invitation to publish in a special issue of the *Health Information Science and Systems* journal.
- Structured and standardized the dataset and training assets to establish a reproducible data pipeline for efficient model development and experimentation.
- Investigated approaches for cross-domain generalization and representation learning across heterogeneous medical imaging datasets.
- Developed visual and presentation assets used to communicate the project during the competition and dissemination stages.

TE AI CUP 2025 – International 3rd Place

Project: AI Generated Machine Downtime Insights
January – May 2025

UI Developer

- Designed and developed the UI/UX for an AI-based industrial downtime analysis platform, translating complex machine time-series data and unsupervised learning outputs into clear, actionable visual insights.
- Developed an accessible and user-centered interface for non-technical users, simplifying the exploration and interpretation of AI-generated insights without requiring prior knowledge of machine learning.
- Defined user flows, information hierarchy, and interaction patterns to improve navigation and facilitate the transition from high-level downtime trends to detailed analytical results.
- Collaborated with the data science and engineering teams to align the interface with the underlying analytical models and ensure that visual representations accurately reflected model outputs.